GP AI: Precision Care for Diabates using Graph RAG

Jad Mokdad, Mehul Goenka, Huu Khue Pham, Sriya Shabadu, Goyeun Yun

Introduction

Problem Statement

- Current diabetes care predominantly relies on population-level guidelines that overlook patient-specific heterogeneity in genetics, history, and lifestyle.
- Traditional RAG frameworks are unable to capture deep, structured relationships—e.g. between genes, pathways, drugs, lab results, and outcomes.

Why It Matters?

- Our Graph RAG LLM system can enhance patient-specific diabetes recommendations—minimizing adverse events, reducing long-term healthcare burden, and fostering clinician trust through transparent, evidence-backed reasoning.

Our Approach

What is it?

- We present an interactive Graph-RAG system for precision diabetes care that integrates ADA guidelines and drug data into a dynamic knowledge graph. Users interact via a chat interface, where an LLM processes each query to detect intent, retrieve relevant graph data, and deliver medically grounded responses for more precise recommendations in diabetes care.

How it works?

- **Data Integration**
  We ingest ADA guideline and drug label data from FDA into a unified biomedical ontology.
- **Knowledge Graph Construction**
  A dynamic, directed graph links drugs, conditions, lab tests, and protocol rules via edges that denote explicitly defined clinical relationships.
- **Query Handling**
  Users type questions in a chat interface; an LLM classifies intent and extracts clinical entities.
- **Contextual Retrieval**
  The system matches query embeddings and entities against the graph to isolate the most relevant subgraph.
- **Evidence-Backed Generation**
  Retrieved subgraph snippets plus source metadata feed into the LLM, which produces concise, medically grounded answers with inline citations.
- **Interactive Visualization**
  Answers appear alongside an explorable node-link diagram, letting clinicians trace each recommendation back to its original source.

Why It Works?

- Focused subgraph retrieval filters out irrelevant information, improving answer precision.
- Explicit provenance tracing provides transparent reasoning that fosters clinician trust.

Novel Contributions

- Combined semantic similarity metrics capture richer biomedical relationships than flat text retrieval.
- An interactive provenance UI empowers users to explore and validate each recommendation's data sources.

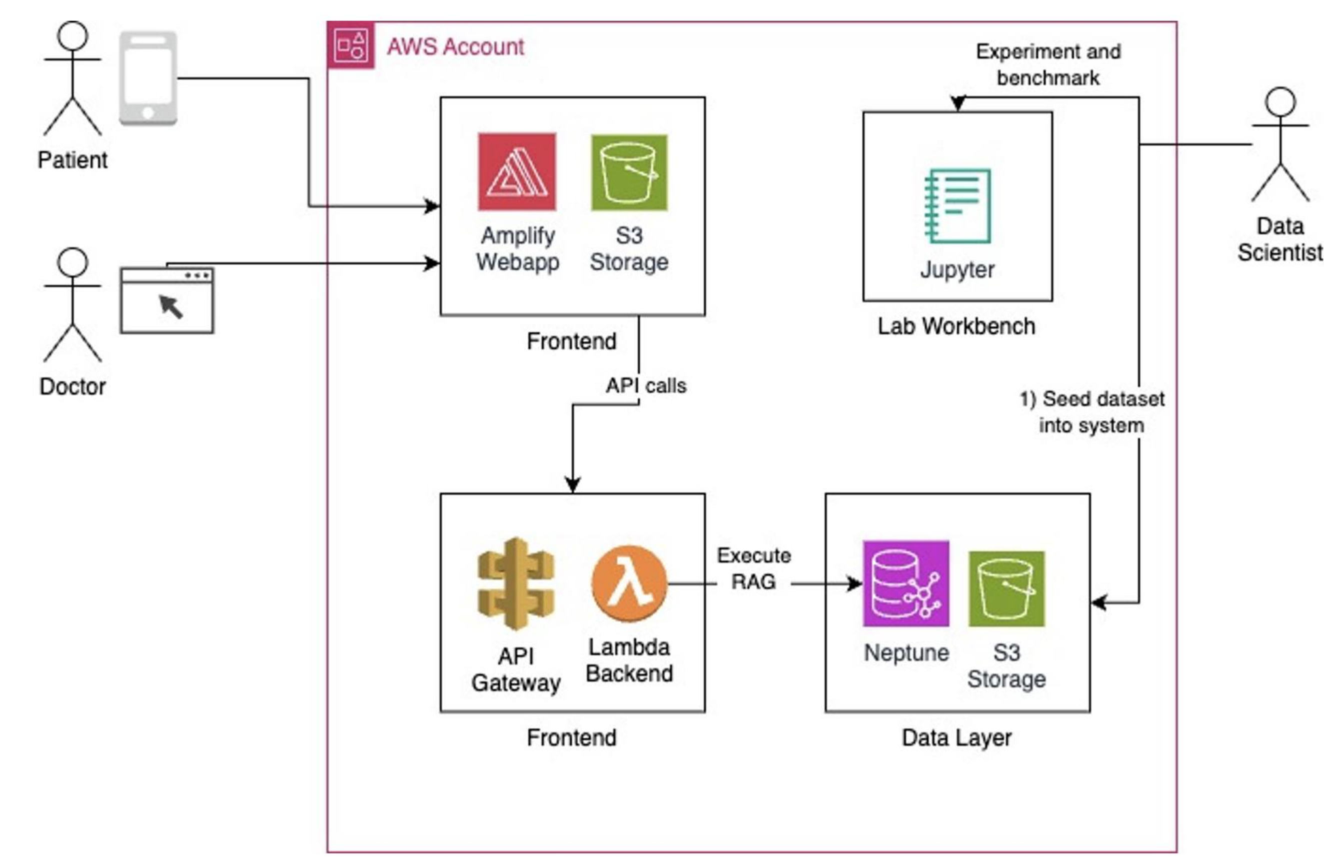

Fig.1 Architecture on AWS

Data

How did we get it and What are its characteristics?

- **ADA guideline:** dataset was sourced from PDF documents (25 MB) and processed into 6,823 individual text chunks using a custom pipeline. Each chunk contains extracted medical intents and entities relevant to diabetes care, and then clinical recommendations in structured JSON format were captured.
- **The drug label:** Originally 7.99 GB in JSON format, was filtered to include 57 diabetes-related drugs. A structured set of drug-related entities (e.g., mechanisms, indications, contraindications) was extracted for graph construction.
- Combined, a total of **5,488 nodes** and **39,929 edges** were created.

Experiments and Results

How did we evaluate our approach?

- We ran GP AI on 80 MRCP SCE Endocrinology & Diabetes sample questions (excluding 3 image-based items), repeating the test five times to compute average accuracy.

What are the results?

- GP AI achieved 65.5% average accuracy—surpassing the MRCP passing mark of 62.5%.

How does our method compare to other methods?

- Compared to baseline LLMs without graph-RAG, GP AI adds traceability without sacrificing performance (see Figure 2).

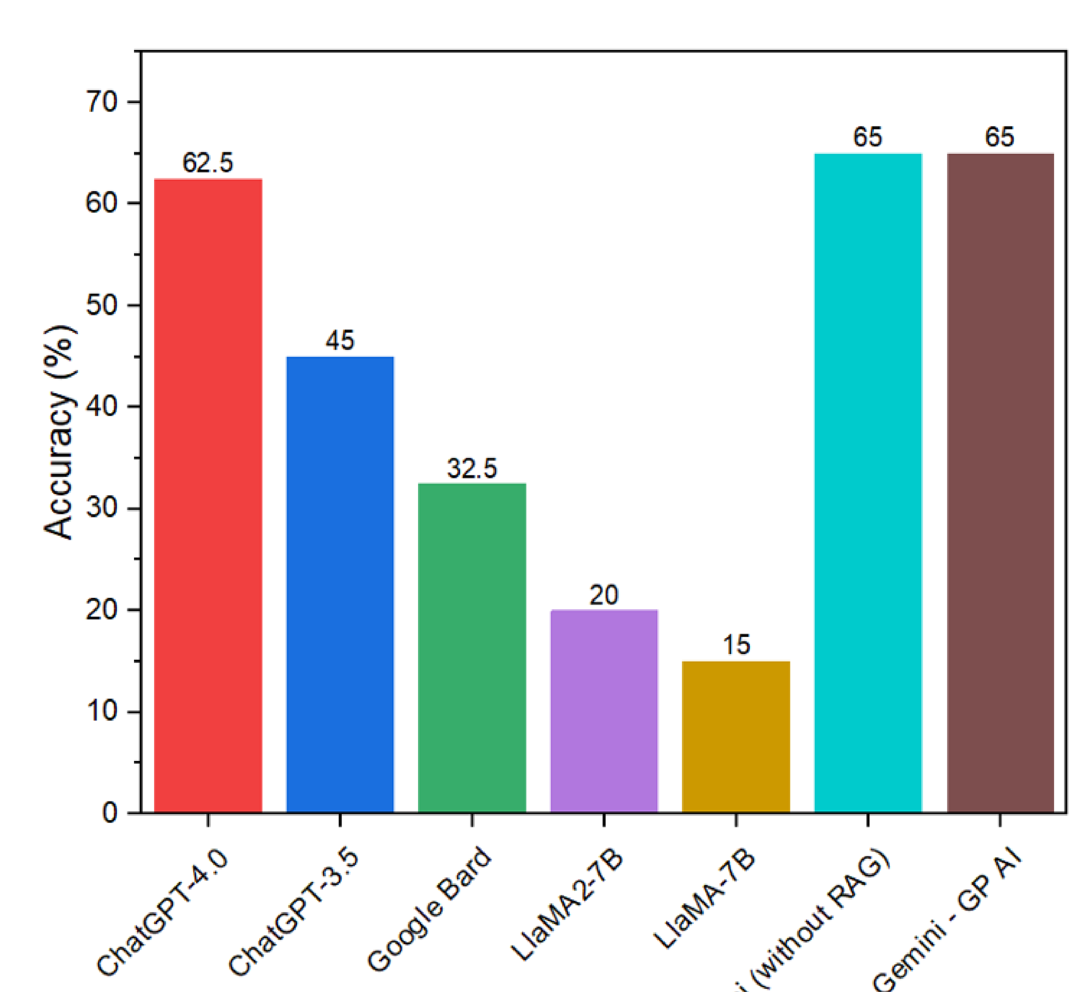

Fig 2. Accuracy Rate on Specialty Certificate Examination